AI-Based Early Malnutrition Prediction and Decision Support System

Introduction

- Malnutrition is a major health problem among children under five years of age.
- It can lead to stunting, wasting, underweight conditions, poor immunity, and delayed growth.
- Early detection of malnutrition is important to prevent severe health complications.
- ICDS and Anganwadi centers collect child health data, but most systems focus only on recording information.
- Existing systems often identify malnutrition after the condition becomes severe.
- Artificial Intelligence can help predict malnutrition risk at an early stage and support timely intervention.

Problem Statement

- Existing child monitoring systems mainly focus on data collection and reporting.
- There is limited support for early risk prediction and decision making.
- ICDS workers find it difficult to identify children who need immediate attention.
- Lack of personalized nutrition guidance and prioritization mechanisms.
- Need for an intelligent system that can assist workers in preventing malnutrition rather than only monitoring it.

Proposed Work

- Collect child health data such as age, weight, height, MUAC, and nutrition history.
- Use AI and Machine Learning algorithms to predict malnutrition risk.
- Classify children into Low Risk, Medium Risk, and High Risk categories.
- Apply SHAP-based Explainable AI to identify factors influencing risk prediction.
- Generate personalized nutrition recommendations based on the child's condition.
- Create a priority ranking system for identifying vulnerable children.
- Develop an interactive dashboard for ICDS workers to monitor child health and growth.

Expected Outcome

- Early identification of children at risk of malnutrition.
- Improved intervention planning by ICDS workers.
- Better monitoring of child growth and nutritional status.
- Personalized nutrition guidance for vulnerable children.
- Efficient prioritization of high-risk children.
- Reduced chances of severe malnutrition through timely action.
- Enhanced decision-making using AI-driven insights.

Novelty

- AI-based early malnutrition risk prediction.
- Explainable AI using SHAP for transparent predictions.
- Personalized nutrition recommendation engine.
- Child priority ranking system for ICDS workers.
- Integrated dashboard for monitoring and decision support.
- Focus on prevention rather than post-detection monitoring.

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